

[BUG] Incomplete Operator Name Mapping for Controlled RX Gate

<https://github.com/PennyLaneAI/pennylane/issues/4660>

ROW 76

[BUG] Incomplete Operator Name Mapping for Controlled RX Gate #4660

New issue



Closed

#4697



c-3543 opened on Oct 7, 2023

...

Expected behavior

Great question. The way operators get translated from PL to Qiskit is via a dictionary lookup given the operator's name in PL (see [here](#) for the source code). The controlled RX gate has a different name in PennyLane depending on how you define it:

```
op = qml.CRX(0.1, wires=[0, 1])
op.name
'CRX'
op = qml.ctrl(qml.RX(0.1, 0), control=[1])
op.name
'C(RX)'
```

Actual behavior

There is a mapping for when the name is CRX, but not for C(RX)

Additional information

Info here:

<https://discuss.pennylane.ai/t/pennylane-ibmq-conversion-error/3498>

Source code

```
import pennylane as qml
from pennylane import numpy as np

property_prices = [4, 8, 6, 3, 12, 15] # total 48
variables_wires = [0, 1, 2, 3, 4, 5]

aux_oracle_wires = [6, 7, 8, 9, 10, 11]

def oracle(variables_wires, aux_oracle_wires):

    def add_k_fourier(k, wires):
        for j in range(len(wires)):
            qml.RZ(k * np.pi / (2**j), wires=wires[j])

    def value_second_sibling():

        qml.QFT(wires = aux_oracle_wires)

        for wire in variables_wires:
            qml.ctrl(add_k_fourier, control = wire)(property_prices[wire], wires = aux_oracle_wires)

        qml.adjoint(qml.QFT)(wires = aux_oracle_wires)

    value_second_sibling()
    qml.FlipSign(sum(property_prices) // 2, wires = aux_oracle_wires)
    qml.adjoint(value_second_sibling)()

dev = qml.device('qiskit.ibmq', wires=12, backend='ibmq_qasm_simulator')

@qml.qnode(dev)
def circuit():

    # step 1
    for wire in variables_wires:
        qml.Hadamard(wires = wire)

    # step 2
    oracle(variables_wires, aux_oracle_wires)

    # step 3
    qml.GroverOperator(wires = variables_wires)

    return qml.probs(wires = variables_wires)

import matplotlib.pyplot as plt
```

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

🔗 **Decompositions for multi-controlled single qubit special unitary ops**
PennyLaneAI/pennylane

Notifications

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Participants



Visual S

```
values = circuit()
plt.bar(range(len(values)), values)
```

Tracebacks

Put full error message here



```
-----
KeyError                                Traceback (most recent call last)
Cell In[5], line 3
      1 import matplotlib.pyplot as plt
----> 3 values = circuit()
      4 plt.bar(range(len(values)), values)

File ~\anaconda3\Lib\site-packages\pennylane\qnode.py:989, in QNode.__call__(self, *args, **kwargs)
    986     self.execute_kwargs.pop("mode")
    988 # pylint: disable=unexpected-keyword-arg
--> 989 res = qml.execute(
    990     (self._tape,),
    991     device=self.device,
    992     gradient_fn=self.gradient_fn,
    993     interface=self.interface,
    994     transform_program=self.transform_program,
    995     gradient_kwargs=self.gradient_kwargs,
    996     override_shots=override_shots,
    997     **self.execute_kwargs,
    998 )
   1000 res = res[0]
   1002 # convert result to the interface in case the qfunc has no parameters

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:757, in execute(tapes, device, gradient_fn,
    754     raise ValueError("Gradient transforms cannot be used with grad_on_execution=True")
    756 ml_boundary_execute = _get_ml_boundary_execute(interface, _grad_on_execution)
--> 757 results = ml_boundary_execute(
    758     tapes, device, execute_fn, gradient_fn, gradient_kwargs, _n=1, max_diff=max_diff
    759 )
    761 results = batch_fn(results)
    762 return program_post_processing(results)

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\autograd.py:317, in execute(tapes, device, execute_fn,
    312 # pylint misidentifies autograd.builtins as a dict
    313 # pylint: disable=no-member
    314 parameters = autograd.builtins.tuple(
    315     [autograd.builtins.list(t.get_parameters()) for t in tapes]
    316 )
--> 317 return _execute(
    318     parameters,
    319     tapes=tapes,
    320     device=device,
    321     execute_fn=execute_fn,
    322     gradient_fn=gradient_fn,
    323     gradient_kwargs=gradient_kwargs,
    324     _n=_n,
    325     max_diff=max_diff,
    326 )[0]

File ~\anaconda3\Lib\site-packages\autograd\tracer.py:48, in primitive.<locals>.f_wrapped(*args, **kwargs)
     46     return new_box(ans, trace, node)
     47 else:
--> 48     return f_raw(*args, **kwargs)

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\autograd.py:378, in _execute(parameters, tapes, device,
    360 if logger.isEnabledFor(logging.DEBUG):
    361     logger.debug(
    362         "Entry with args=(parameters=%s, tapes=%s, device=%s, execute_fn=%s, gradient_fn=%s, gradient_
    363         parameters,
    364         (...))
    365         ":%s".join(str(i) for i in inspect.getouterframes(inspect.currentframe(), 2)[1][1:3]),
    366     )
--> 378 res, jacs = execute_fn(tapes, **gradient_kwargs)
    380 return res, jacs

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:623, in execute.<locals>.inner_execute_with_
    622 def inner_execute_with_empty_jac(tapes, **_):
--> 623     return (inner_execute(tapes), [])

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:255, in _make_inner_execute.<locals>.inne
    253 if numpy_only:
    254     tapes = tuple(qml.transforms.convert_to_numpy_parameters(t) for t in tapes)
--> 255 return cached_device_execution(tapes)
```

```

File ~\anaconda3\Lib\site-packages\pennylane\interfaces\execution.py:377, in cache_execute.<locals>.wrapper(tc
372         return (res, []) if return_tuple else res
374 else:
375     # execute all unique tapes that do not exist in the cache
376     # convert to list as new device interface returns a tuple
--> 377     res = list(fn(tuple(execution_tapes.values()), **kwargs))
379 final_res = []
381 for i, tape in enumerate(tapes):

File ~\anaconda3\Lib\contextlib.py:81, in ContextDecorator.__call__.<locals>.inner(*args, **kwargs)
78 @wraps(func)
79 def inner(*args, **kwargs):
80     with self._recreate_cm():
--> 81         return func(*args, **kwargs)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\ibmq.py:91, in IBMQDevice.batch_execute(self, circuits)
90 def batch_execute(self, circuits): # pragma: no cover, pylint:disable=arguments-differ
--> 91     res = super().batch_execute(circuits, timeout=self.timeout_secs)
92     if self.tracker.active:
93         self._track_run()

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:473, in QiskitDevice.batch_execute(self,
470 def batch_execute(self, circuits, timeout: int = None):
471     # pylint: disable=missing-function-docstring
--> 473     compiled_circuits = self.compile_circuits(circuits)
475     # Send the batch of circuit objects using backend.run
476     self._current_job = self.backend.run(compiled_circuits, shots=self.shots, **self.run_args)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:462, in QiskitDevice.compile_circuits(sel
458 for circuit in circuits:
459     # We need to reset the device here, else it will
460     # not start the next computation in the zero state
461     self.reset()
--> 462     self.create_circuit_object(circuit.operations, rotations=circuit.diagonalizing_gates)
464     compiled_circ = self.compile()
465     compiled_circ.name = f"circ{len(compiled_circuits)}"

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:273, in QiskitDevice.create_circuit_objec
261 """Builds the circuit objects based on the operations and measurements
262 specified to apply.
263 (...)
269         pre-measurement into the eigenbasis of the observables.
270 """
271 rotations = kwargs.get("rotations", [])
--> 273 applied_operations = self.apply_operations(operations)
275 # Rotating the state for measurement in the computational basis
276 rotation_circuits = self.apply_operations(rotations)

File ~\anaconda3\Lib\site-packages\pennylane_qiskit\qiskit_device.py:322, in QiskitDevice.apply_operations(sel
318     par[idx] = p.tolist()
320 operation = operation.name
--> 322 mapped_operation = self._operation_map[operation]
324 self.qubit_state_vector_check(operation)
326 qregs = [self._reg[i] for i in device_wires.labels]

KeyError: 'C(RZ)'

```

System information

```

Name: PennyLane
Version: 0.32.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: GitHub - PennyLaneAI/pennylane: PennyLane is a cross-platform Python library for differentiable pro
Author:
Author-email:
License: Apache License 2.0
Location: C:\Users\Carter\anaconda3\Lib\site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning, PennyLane-qiskit

Platform info: Windows-10-10.0.22621-SP0
Python version: 3.11.5
Numpy version: 1.23.5
Scipy version: 1.11.1
Installed devices:

default.gaussian (PennyLane-0.32.0)
default.mixed (PennyLane-0.32.0)
default.qubit (PennyLane-0.32.0)
default.qubit.autograd (PennyLane-0.32.0)

```

```
default.qubit.cirq (PennyLane-0.32.0)  
default.qubit.jax (PennyLane-0.32.0)  
default.qubit.tf (PennyLane-0.32.0)  
default.qubit.torch (PennyLane-0.32.0)  
default.qutrit (PennyLane-0.32.0)  
null.qubit (PennyLane-0.32.0)  
lightning.qubit (PennyLane-Lightning-0.32.0)  
qiskit.aer (PennyLane-qiskit-0.32.0)  
qiskit.basicaer (PennyLane-qiskit-0.32.0)  
qiskit.ibmq (PennyLane-qiskit-0.32.0)  
qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.32.0)  
qiskit.ibmq.sampler (PennyLane-qiskit-0.32.0)  
qiskit.remote (PennyLane-qiskit-0.32.0)
```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



c-3543 added **bug** on Oct 7, 2023



isaacdevlugt on Oct 11, 2023

Contributor ...

Hey @c-3543! Thanks for making the bug report. Just for the paper trail, this came up as an issue in the PL-qiskit plugin because `QISKIT_OPERATION_MAP_SELF_ADJOINT` doesn't have a `'C(RZ)'` key but does have a `'CRZ'` one.

https://github.com/PennyLaneAI/pennylane-qiskit/blob/d6964c723278272196c9fdea57bd2da4db9344e9/pennylane_qiskit/qiskit_device.py#L36



albi3ro mentioned this on Oct 19, 2023

[Decompositions for multi-controlled single qubit special unitary ops #4697](#)

albi3ro closed this as **completed** in [#4697](#) on Oct 23, 2023

albi3ro added a commit that references this issue on Oct 23, 2023

[Decompositions for multi-controlled single qubit special unitary ops](#) ...

Verified a7114a9

Alex-Preciado changed the title **[BUG] [BUG] Incomplete Operator Name Mapping for Controlled RX Gate** on Nov 12, 2023

Decompositions for multi-controlled single qubit special unitary ops #4697

<> Code

Merged albi3ro merged 4 commits into master from multi-controlled-special-unitary on Oct 23, 2023

Conversation 9 Commits 4 Checks 0 Files changed 3

+56 -12



albi3ro commented on Oct 19, 2023

Contributor

[sc-43590] Fixes #4473 Fixes #4660

We previously turned off automatic decompositions for multicontrolled single qubit gates as our decompositions only applied to special unitary operations.

In this PR, I just check if the matrix is special unitary using its determinant. Since this is only for single qubit gates, I'm not concerned about the performance of this.

To decompose non-special unitary gates like `PauliZ`, we would need to include a controlled global phase operation. With the addition of the `GlobalPhase`, we can do so in a follow-up PR. But this one solves the immediate problems.

2

multicontrolled decompositions

3fdfa57

albi3ro requested a review from a team 2 years ago



codecov bot commented on Oct 19, 2023 • edited

Codecov Report

All modified and coverable lines are covered by tests

Comparison is base (39c996a) 0.00% compared to head (6454026) 99.63%.

Additional details and impacted files

View full report in Codecov by Sentry.

Have feedback on the report? Share it here.

Reviewers

mudit2812

timmysilv

lillian542

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

v0.33

Development

Successfully merging this pull request may close these issues.

[BUG] Incomplete Operator Name Mapping f...

[BUG] Decomposition of multi-controlled RZ ...

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